

Chapter 2: Complex Analysis

Special Functions of Mathematical Physics: A Tourist's Guidebook

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Outline

- 1 Taylor series representation
- 2 Complex variables
- 3 Derivatives
- 4 Path integrals
- 5 Taylor and Laurent series
- 6 Singularities
- 7 Residues
- 8 Residue calculus
- 9 Method of steepest descent

Where we are going

To analyze the special functions of physics we first build the tools of *complex analysis*.

- **Taylor and Laurent series** represent a function near a point, even near a singularity.
- **Analytic functions** (those with a derivative in the complex sense) obey the *Cauchy–Riemann conditions* and are infinitely differentiable.
- **Cauchy's integral theorem** and the **residue theorem** turn contour integrals into a count of poles, giving a powerful machine for evaluating real integrals and infinite sums.
- The **method of steepest descent** extracts the large-parameter behavior of integrals—the asymptotics we will need for the named functions.

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Matching a polynomial to a function

We want the best local polynomial approximation

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n$$

to a smooth function $f(x)$ near $x = 0$. Demand that p and its derivatives agree with those of f at the origin: $f(0) = p(0)$, $f'(0) = p'(0)$, $f''(0) = p''(0)$, and so on.

Differentiating p and evaluating at $x = 0$ strips off one coefficient at a time, because every term with a positive power of x vanishes there:

$$\begin{aligned} p(0) = a_0 &\Rightarrow a_0 = f(0), & p'(0) = a_1 &\Rightarrow a_1 = f'(0), \\ p''(0) = 2a_2 &\Rightarrow a_2 = \frac{1}{2}f''(0), & p^{(3)}(0) = 6a_3 &\Rightarrow a_3 = \frac{1}{6}f^{(3)}(0), \end{aligned}$$

and in general $p^{(n)}(0) = n! a_n$ gives

$$a_n = \frac{1}{n!} f^{(n)}(0). \quad (2.1.1)$$

Here $n! = n \cdot (n-1) \cdots 2 \cdot 1$ is the factorial.

Matching a polynomial to a function (cont.)

Collecting the coefficients gives the *Taylor polynomial*. Letting the number of terms $n \rightarrow \infty$ gives the *formal Taylor series*; when f is analytic, this series converges back to f in its interval (or disk) of convergence:

$$f(x) = f(0) + xf'(0) + \frac{1}{2}x^2f''(0) + \frac{1}{3!}x^3f'''(0) + \cdots, \quad (2.1.2)$$

for x in that convergence region. About an arbitrary point x_0 ,

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2}(x - x_0)^2f''(x_0) + \frac{1}{3!}(x - x_0)^3f'''(x_0) + \cdots. \quad (2.1.3)$$

A function that has a Taylor series representation at x_0 is said to be *analytic* at x_0 .

Examples: the elementary power series

Exponential. Define e^x by $\frac{d}{dx}e^x = e^x$ with $e^0 = 1$. Then $f^{(n)}(0) = 1$ for all n , so (2.1.2) gives

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \cdots. \quad (2.1.4)$$

Since $n!$ grows faster than x^n , the series converges for all x .

Cosine and sine. Using $\frac{d}{dx}\cos x = -\sin x$ and $\frac{d}{dx}\sin x = \cos x$, together with $\sin 0 = 0$, $\cos 0 = 1$,

$$\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{4!}x^4 - \cdots, \quad (2.1.5)$$

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots. \quad (2.1.6)$$

Both converge for all x .

Detour: arithmetic as repeated operations

The book's arithmetic detour puts familiar operations into one ladder:

succession $\longrightarrow$ addition $\longrightarrow$ multiplication $\longrightarrow$ exponentiation $\longrightarrow$ tetration.

Each operation repeats the previous one:

$$x + y = \underbrace{S(S(\cdots S(x)))}_{y \text{ times}}, \quad xy = \underbrace{x + x + \cdots + x}_{y \text{ times}}, \quad x^y = \underbrace{x \cdot x \cdots x}_{y \text{ times}}.$$

The next operation, *tetration*, repeats exponentiation. Let $T_1(x) = x$ and $T_{y+1}(x) = x^{T_y(x)}$. Then

$$T_1(3) = 3, \quad T_2(3) = 3^3 = 27, \quad T_3(3) = 3^{27} = 7,625,597,484,987.$$

The book also writes tetration as $x \uparrow\uparrow y$, or as a left superscript in some older notation. The next value

$$T_4(3) = 3^{3^{27}}$$

has

$$\lfloor 3^{27} \log_{10} 3 \rfloor + 1 \approx 3.64 \times 10^{12}$$

decimal digits. This is the point of the detour: iterated operations quickly outgrow ordinary notation and force us to invent better symbolic machinery.

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Imaginary unit, Cartesian and polar form

Define the imaginary unit i by $i^2 = -1$, i.e. $i = \sqrt{-1}$. A *complex number* is a linear combination of real and imaginary parts. Powers of i cycle:

$$\{i^0, i^1, i^2, i^3, i^4, \dots\} = \{1, i, -1, -i, 1, \dots\}.$$

In *Cartesian* form $z = x + iy$; we call $x = \operatorname{Re} z$ the real part and $y = \operatorname{Im} z$ the imaginary part.

Put $x = i\theta$ in the exponential series (2.1.4). Each power of i follows the cycle $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, so

$$e^{i\theta} = 1 + i\theta + \frac{1}{2}(i\theta)^2 + \frac{1}{3!}(i\theta)^3 + \frac{1}{4!}(i\theta)^4 + \dots = 1 + i\theta - \frac{1}{2}\theta^2 - i\frac{1}{3!}\theta^3 + \frac{1}{4!}\theta^4 + \dots.$$

The even powers are real and the odd powers carry a factor i ; grouping them and comparing with (2.1.5)–(2.1.6) gives *Euler's formula*:

$$e^{i\theta} = \left(1 - \frac{1}{2}\theta^2 + \frac{1}{4!}\theta^4 - \dots\right) + i\left(\theta - \frac{1}{3!}\theta^3 + \frac{1}{5!}\theta^5 - \dots\right) = \cos \theta + i \sin \theta. \quad (2.2.1)$$

Imaginary unit, Cartesian and polar form (cont.)

Euler's formula gives the *polar* form of a complex number,

$$z = r e^{i\theta}, \quad (2.2.2)$$

where $r = |z|$ is the *modulus* (magnitude) and $\theta = \arg z$ is the *argument* (phase). From $z = x + iy$,

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arg z,$$

where $\arg z$ means the quadrant-aware angle (often computed as $\text{atan2}(y, x)$), not the single-valued real arctangent alone. Multiplication scales by the moduli and adds the arguments: if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ then $z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$.

The *complex conjugate* of $z = r e^{i\theta}$ is $z^* = r e^{-i\theta}$ (in Cartesian form $(x + iy)^* = x - iy$), so that

$$z z^* = (r e^{i\theta}) (r e^{-i\theta}) = r^2 = |z|^2.$$

The conjugate also recovers the Cartesian components:

$$\operatorname{Re} z = \frac{z + z^*}{2}, \quad \operatorname{Im} z = \frac{z - z^*}{2i}. \quad (2.2.3)$$

Imaginary unit, Cartesian and polar form (cont.)

From Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$ and (replacing θ by $-\theta$, since cosine is even and sine is odd) $e^{-i\theta} = \cos \theta - i \sin \theta$. Adding the two cancels the sines, while subtracting cancels the cosines:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (2.2.4)$$

We will use these identities throughout.

Numerical derivatives and the complex step

A truncated Taylor series with a small step h estimates a derivative. From

$$f(x+h) = f(x) + hf'(x) + \frac{1}{2}h^2f''(x) + \frac{1}{6}h^3f'''(x) + \cdots,$$

solving for $f'(x)$ gives us

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{1}{2}hf''(x) - \frac{1}{6}h^2f'''(x) + \cdots, \quad (2.2.5)$$

so the forward difference is first order: $f'(x) = \frac{f(x+h) - f(x)}{h} + O(h)$.

For the centered difference, write both Taylor expansions:

$$f(x+h) = f(x) + hf'(x) + \frac{1}{2}h^2f''(x) + \frac{1}{6}h^3f'''(x) + \cdots,$$

$$f(x-h) = f(x) - hf'(x) + \frac{1}{2}h^2f''(x) - \frac{1}{6}h^3f'''(x) + \cdots.$$

Subtracting cancels the even terms:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{1}{3}h^3f'''(x) + \cdots.$$

Solving for $f'(x)$ gives the *centered difference*, which is second order:

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} - \frac{1}{6}h^2f'''(x) + \cdots = \frac{f(x+h) - f(x-h)}{2h} + O(h^2). \quad (2.2.6)$$

Numerical derivatives and the complex step (cont.)

In floating point, $f(x+h) - f(x)$ is corrupted by round-off of size $|f(x)|\varepsilon$ (machine epsilon $\varepsilon \approx 10^{-16}$), so the forward difference carries two errors:

$$f'(x) = \frac{f(x+h) - f(x)}{h} \underbrace{+ O(h)}_{\text{truncation}} + \underbrace{O\left(\frac{|f(x)|\varepsilon}{h}\right)}_{\text{round-off}}.$$

For a real, analytic f we can sidestep the round-off by stepping in the imaginary direction. From

$$f(x+ih) = f(x) + ihf'(x) - \frac{1}{2}h^2f''(x) - i\frac{1}{6}h^3f'''(x) + \cdots,$$

taking the imaginary part gives $\text{Im}[f(x+ih)] = hf'(x) - \frac{1}{6}h^3f'''(x) + \cdots$. Solving for $f'(x)$ gives us

$$f'(x) = \frac{\text{Im}[f(x+ih)]}{h} + \frac{1}{6}h^2f'''(x) + \cdots = \frac{\text{Im}[f(x+ih)]}{h} + O(h^2). \quad (2.2.7)$$

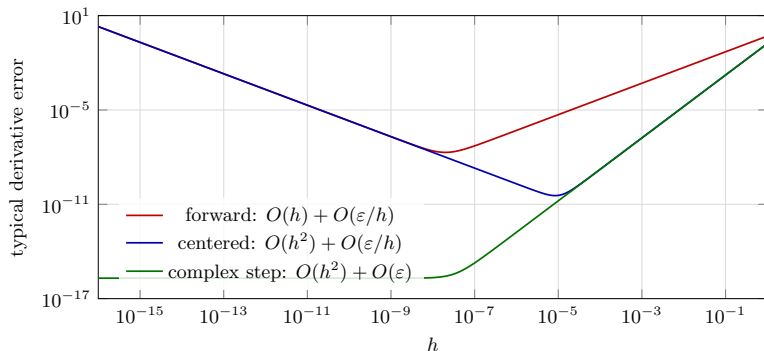
No subtraction of nearly equal numbers occurs, so this *complex-step* derivative is not limited by round-off error.

Derivative error as the step size changes

The Taylor expansion predicts the slopes of the error curves. For a typical smooth function in double precision,

$$E_{\text{forward}}(h) \sim Ch + \frac{C'\varepsilon}{h}, \quad E_{\text{centered}}(h) \sim Ch^2 + \frac{C'\varepsilon}{h}, \quad E_{\text{complex}}(h) \sim Ch^2 + C'\varepsilon.$$

The complex-step curve avoids the $1/h$ round-off wall because no nearly equal real numbers are subtracted.



Multivalued functions and branch cuts

A function of a real variable is easy to graph; its inverse may be *multivalued*. For $f(x) = x^2$ the inverse has two branches, $f^{-1}(y) = \pm\sqrt{y}$ for $y \geq 0$. We pick a *principal branch*, usually $f^{-1}(y) = +\sqrt{y}$, to make it single-valued.

The arcsine is also multivalued: every $x = n\pi$ maps to $y = 0$ (more generally x , $x + 2\pi n$, and $\pi - x$ share the same sine). Since $\sin n\pi = 0$ for all integers n , $\sin^{-1} 0 = n\pi$ for all integer n . We restrict to the principal branch $x \in [-\pi/2, \pi/2]$.

Multivalued functions and branch cuts (cont.)

The logarithm. As the inverse of the exponential, $\log z$ is multivalued. Writing $z = re^{i\theta}$ and inserting the ambiguity $e^{i2\pi n} = 1$,

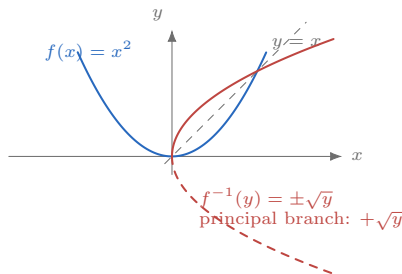
$$\begin{aligned}\log z &= \log r + \log e^{i\theta} = \log r + \log e^{i(\theta+2\pi n)} \\ &= \log r + i(\theta + 2\pi n) = \log |z| + i \arg z,\end{aligned}$$

where $\arg z = \theta + 2\pi n$. The principal branch takes $-\pi < \text{Arg } z < \pi$:

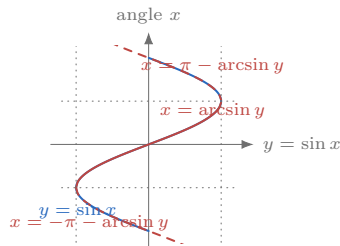
$\text{Log } z = \log |z| + i \text{Arg } z$ with $\text{Arg } z$ chosen in that interval. To keep a multivalued function single-valued we make a *branch cut* in the plane. A natural cut for $\sqrt{z}$ or $\log z$ runs along the negative real axis, but any simple path from the branch point to infinity can be used if it keeps the important region continuous. Crossing the cut, $\log z$ jumps by $2\pi i$.

Real graphs and inverse branches

real function and inverse branches



$\sin x$, $\arcsin$, and branch choices



principal branch uses $-\pi/2 \leq x \leq \pi/2$

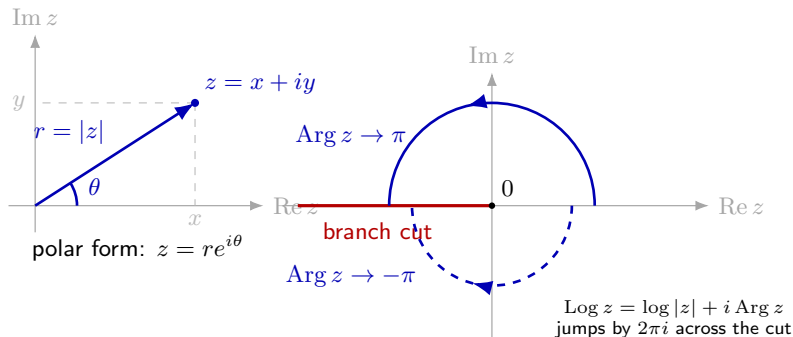
The book first motivates branch choices using real inverse functions: x^2 has two inverse branches, and $\sin x$ repeats values infinitely often. Complex roots and logarithms behave the same way, except the branch choice is made in the plane.

Complex-variable pictures and branch cuts

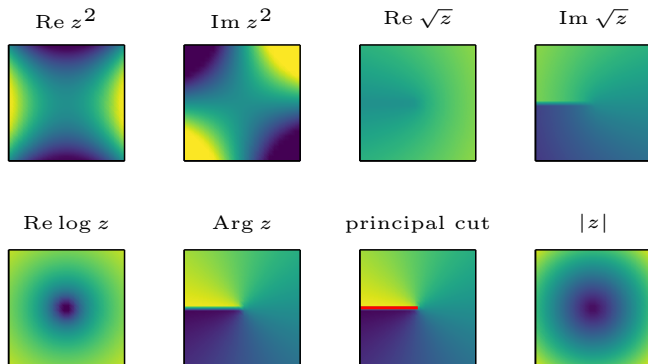
A complex-valued function has two real inputs and two real outputs:

$$z = x + iy, \quad f(z) = u(x, y) + iv(x, y).$$

Instead of one graph, we often inspect level curves, real and imaginary parts, or how a chosen branch changes around a singular point.



Component plots for complex functions



For $f(z) = u(x, y) + iv(x, y)$, a single real graph is replaced by component plots. The gallery shows the real and imaginary parts of z^2 , the principal square root, and the principal logarithm; the jump in $\operatorname{Arg} z$ is exactly the branch cut.

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The complex derivative and Cauchy–Riemann

The derivative of $f(z)$ is

$$f'(z) = \frac{df}{dz} = \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z}, \quad (2.3.1)$$

provided the limit is the same for *all* directions δz . If the derivative exists only at z_0 , f is complex differentiable there. If it exists in a neighborhood of z_0 , f is *analytic* at z_0 ; if it exists everywhere in $\mathbb{C}$, f is *entire*; where it fails, z_0 is a *singular point*.

Write $z = x + iy$ and $f = u(x, y) + iv(x, y)$. Approaching along the real axis ($\delta z = \delta x$, $\delta y = 0$):

$$\frac{df}{dz} = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}.$$

Approaching along the imaginary axis ($\delta z = i \delta y$, $\delta x = 0$) divides by the extra factor i ; using $1/i = -i$,

$$\frac{df}{dz} = \frac{1}{i} \frac{\partial f}{\partial y} = -i \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}.$$

The complex derivative and Cauchy–Riemann (cont.)

For the derivative to exist these must agree. Matching real and imaginary parts gives the *Cauchy–Riemann conditions*:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \quad (2.3.2)$$

Equivalently, $i \partial f / \partial x = \partial f / \partial y$. These conditions are necessary for analyticity. If the first partial derivatives are continuous in a neighborhood, they are also sufficient, so in that common smooth case $f = u + iv$ is analytic if and only if it satisfies the Cauchy–Riemann conditions.

Examples.

- $z^2 = (x^2 - y^2) + i 2xy$ is entire: $u_x = 2x = v_y$ and $u_y = -2y = -v_x$.
- On a chosen branch, $\log z$ is analytic away from 0 and the branch cut. Locally

$$f(x, y) = \log \sqrt{x^2 + y^2} + i \operatorname{Arg}(x + iy),$$

and

$$\frac{\partial f}{\partial x} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}, \quad \frac{\partial f}{\partial y} = \frac{y}{x^2 + y^2} + i \frac{x}{x^2 + y^2},$$

so $i \partial_x f = \partial_y f$ on that branch.

The complex derivative and Cauchy–Riemann (cont.)

- $1/z$ is analytic except at $z = 0$. With $f = 1/(x + iy)$, $\partial_x f = -1/(x + iy)^2$ and $\partial_y f = -i/(x + iy)^2$, so $i\partial_x f = \partial_y f$ except at the origin, where f blows up.
- $z^* = x - iy$ has $u = x$, $v = -y$, so $u_x = 1$ but $v_y = -1$: the Cauchy–Riemann conditions never hold, z^* is nowhere analytic.

A non-constant polynomial has a genuine *pole at infinity*; other entire functions need not (e.g. e^z has an essential singularity at ∞ , and constants are bounded).

Harmonic functions, potentials, streamlines

Laplace's equation $\nabla^2 u = 0$ governs steady-state temperature, electrostatic potential, and elastic membranes. Its solutions are *harmonic functions*. If $f = u + iv$ is analytic, both u and v are harmonic. From the Cauchy–Riemann conditions,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial y \partial x}, \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial x \partial y},$$

and adding cancels the mixed partials:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \text{similarly } \nabla^2 v = 0. \quad (2.3.3)$$

A nonconstant harmonic function has no strict local maximum or minimum in the interior of its domain. Away from flat critical points, this is why its level curves look saddle-like rather than peak-like or bowl-like.

Harmonic functions, potentials, streamlines (cont.)

In fluid mechanics and electrostatics we call u the *complex potential* and v the *stream function*. Curves $u = \text{const}$ are *equipotentials*, curves $v = \text{const}$ are *streamlines*. Their gradients are orthogonal, since by Cauchy–Riemann

$$\nabla u \cdot \nabla v = \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} = 0.$$

Example (z^2). With $f = z^2 = (x^2 - y^2) + i 2xy$, the potential is $u = x^2 - y^2$ and the stream function is $v = 2xy$. Thus

$$u = c_1 \iff x^2 - y^2 = c_1, \quad v = c_2 \iff 2xy = c_2.$$

These are two orthogonal families of hyperbolas. The orthogonality check is direct: $\nabla u = (2x, -2y)$, $\nabla v = (2y, 2x)$, so $\nabla u \cdot \nabla v = 4xy - 4xy = 0$.

Example ($1/z$). With $f = 1/z$ and $z = x + iy$,

$$f = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}.$$

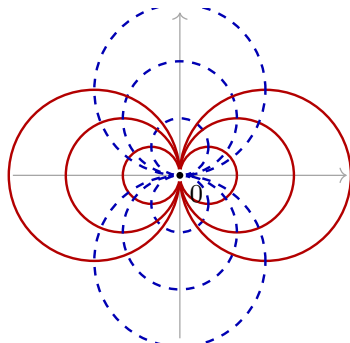
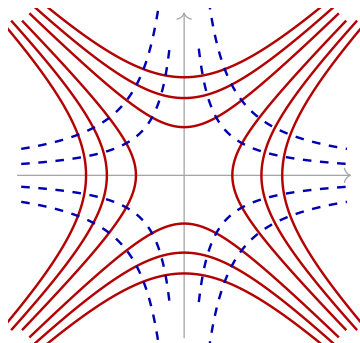
In polar coordinates ($x = r \cos \theta$, $y = r \sin \theta$),

$$u(r, \theta) + iv(r, \theta) = r^{-1} \cos \theta - i r^{-1} \sin \theta,$$

Harmonic functions, potentials, streamlines (cont.)

so $u = r^{-1} \cos \theta$ is the complex potential and $v = -r^{-1} \sin \theta$ is the stream function. The equation $r^{-1} \cos \theta = c$ rearranges to $r = (1/c) \cos \theta$; after renaming $1/c$ as c_1 , the equipotentials are $r = c_1 \cos \theta$. Similarly the streamlines are $r = c_2 \sin \theta$.

Harmonic functions, potentials, streamlines (cont.)



— $u = \text{constant}$ - - - $v = \text{constant}$

The red curves are $u = \text{constant}$ and the dashed blue curves are $v = \text{constant}$. The Cauchy–Riemann equations force the two families to meet orthogonally wherever $f'(z) \neq 0$.

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Contour integrals and a key circle integral

Along the real axis, the usual Riemann integral is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} f(x_j)(x_{j+1} - x_j).$$

For a path γ from $z = c$ to $z = d$, the same Riemann-sum idea defines the *path* (contour, line) integral:

$$\int_{\gamma} f(z) dz = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(z_i)(z_{i+1} - z_i), \quad (2.4.1)$$

with $z_0 = c$, $z_n = d$. If $c = d$ the path is closed, written $\oint_{\gamma}$; by convention closed paths are traversed counterclockwise.

Powers of z on a circle. On $|z| = R$ write $z = Re^{i\theta}$, $dz = iRe^{i\theta} d\theta$. Then

$$\oint_{|z|=R} z^n dz = iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Contour integrals and a key circle integral (cont.)

(For $n = -1$ the integrand is $i d\theta$; for $n \neq -1$ the exponential integrates to a periodic function over a full period.) In general,

$$\oint_{|z-z_0|=R} (z - z_0)^n dz = 2\pi i \delta_{n,-1}. \quad (2.4.2)$$

Contour integrals and a key circle integral (cont.)

Antiderivatives. Parameterize γ from $z = a$ to $z = b$ as $z = \gamma(t)$, $t \in [0, 1]$, with $\gamma(0) = a$ and $\gamma(1) = b$. Let $F' = f$. By the fundamental theorem of calculus,

$$\int_{\gamma} f(z) dz = \int_0^1 F'(\gamma(t))\gamma'(t) dt = F(\gamma(1)) - F(\gamma(0)) = F(b) - F(a).$$

If γ is closed ($a = b$) and F is single-valued, then $\oint_{\gamma} f dz = 0$.

Reworking the circle example over any simple closed γ around the origin:

- $n \neq -1$: $F(z) = z^{n+1}/(n+1)$ is single-valued, so $\oint_{\gamma} z^n dz = 0$.
- $n = -1$: $F(z) = \log z$ has a branch cut. Put the cut on the negative real axis and shorten γ by $\pm\varepsilon$ on each side. On the cut-open curve, use $\log z = \log |z| + i \arg z$: the radius returns to its starting value while the angle runs from $\theta = -\pi + \varepsilon$ to $\theta = \pi - \varepsilon$, so

$$\int_{\gamma} \frac{1}{z} dz = \left[\log r + i\theta \right]_{\theta=-\pi+\varepsilon}^{\theta=\pi-\varepsilon} = i(2\pi - 2\varepsilon) = 2\pi i - 2\varepsilon i.$$

As $\varepsilon \rightarrow 0$, $\oint_{\gamma} \frac{1}{z} dz = 2\pi i$.

Cauchy's integral theorem and formula

Theorem (Cauchy's Integral Theorem)

If $f(z)$ is analytic in a simply connected region and γ is a closed path in that region, then $\oint_{\gamma} f(z) dz = 0$.

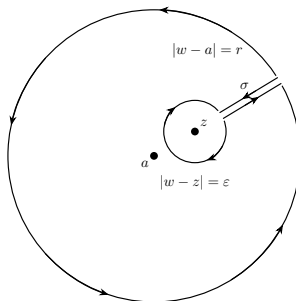
Proof. The following calculation proves the smooth case; the full Cauchy–Goursat theorem removes this extra smoothness assumption. Assume first that $\gamma = \partial D$ is positively oriented, piecewise smooth, and simple. With $f = u + iv$ and $dz = dx + i dy$, Green's theorem and Cauchy–Riemann give

$$\begin{aligned}\oint_{\gamma} f dz &= \oint_{\gamma} (u dx - v dy) + i \oint_{\gamma} (v dx + u dy) \\ &= \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy + i \iint_D \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy = 0,\end{aligned}$$

since each integrand vanishes by (2.3.2). □

Cauchy's integral theorem and formula (cont.)

Cauchy's integral formula. Let f be analytic and γ encircle z_0 . Deform γ to a small circle $z = z_0 + \varepsilon e^{i\theta}$; the two sides of the connecting slit are traversed in opposite directions, so they cancel:



Cauchy's integral theorem and formula (cont.)

On the small circle, $z - z_0 = \varepsilon e^{i\theta}$ and $dz = i\varepsilon e^{i\theta} d\theta$, so the factor $\varepsilon e^{i\theta}$ cancels between numerator and denominator:

$$\oint_{\gamma} \frac{f(z)}{z - z_0} dz = \int_0^{2\pi} \frac{f(z_0 + \varepsilon e^{i\theta})}{\varepsilon e^{i\theta}} i\varepsilon e^{i\theta} d\theta = i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta}) d\theta.$$

As $\varepsilon \rightarrow 0$ continuity gives $f(z_0 + \varepsilon e^{i\theta}) \rightarrow f(z_0)$, a constant that comes out of the integral, and $\int_0^{2\pi} d\theta = 2\pi$:

$$\oint_{\gamma} \frac{f(z)}{z - z_0} dz = if(z_0) \int_0^{2\pi} d\theta = 2\pi i f(z_0). \quad (2.4.3)$$

Differentiating Cauchy's formula with respect to z_0 (equivalently, matching the Taylor coefficient of f at z_0) gives the formula for higher poles and the *derivative formula*. The first formula is for $n = 1, 2, \dots$; the second is for $n = 0, 1, 2, \dots$:

$$\oint_{\gamma} \frac{f(z)}{(z - z_0)^n} dz = \frac{2\pi i}{(n-1)!} f^{(n-1)}(z_0), \quad f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz. \quad (2.4.4)$$

Cauchy's integral theorem and formula (cont.)

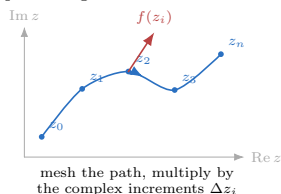
Orientation check. Let

$$I = \oint_{\gamma} \frac{z^2}{z-2} dz.$$

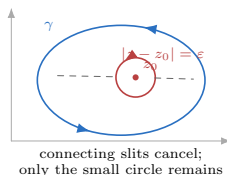
If γ winds counterclockwise once around $z = 2$, Cauchy's formula gives $I = 2\pi i f(2)$ with $f(z) = z^2$, hence $I = 2\pi i \cdot 4 = 8\pi i$. If the same curve is traversed clockwise, the orientation reverses the sign: $I = -8\pi i$. If γ does not enclose $z = 2$, the integrand is analytic inside γ , so Cauchy's theorem gives $I = 0$.

Geometry behind contour arguments

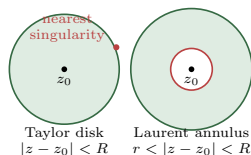
path integral as a Riemann sum



Cauchy formula: shrink to z_0



Taylor disk vs Laurent annulus



The same three pictures recur throughout the chapter: Riemann sums along a path, Cauchy's deformation to a small circle around z_0 , and the disk/annulus distinction between Taylor and Laurent convergence.

Worked example: partial fractions and Cauchy

Evaluate $\oint_{|z|=2} \frac{e^z}{z^2(z-1)} dz$.

Partial fractions: $\frac{1}{z^2(z-1)} = \frac{az+b}{z^2} + \frac{c}{z-1}$. Clearing denominators,

$(az+b)(z-1) + cz^2 = 1$, so $az^2 + bz - az - b + cz^2 = 1$. Matching coefficients,

$$\left. \begin{array}{l} a+c=0 \\ b-a=0 \\ -b=1 \end{array} \right\} \Rightarrow c=1, \quad a=-1, \quad b=-1.$$

Hence the integrand splits as

$$\frac{e^z}{z^2(z-1)} = \frac{e^z}{z-1} - \frac{(z+1)e^z}{z^2}.$$

Worked example: partial fractions and Cauchy (cont.)

The first piece: by Cauchy's integral formula (2.4.3) with $z_0 = 1$ inside $|z| = 2$,

$$\oint_{|z|=2} \frac{e^z}{z-1} dz = 2\pi i e^1.$$

The second piece: set $f(z) = (z+1)e^z$, a double pole at $z = 0$. Then

$$f'(z) = e^z + (z+1)e^z = (z+2)e^z, \quad f'(0) = 2,$$

so by the derivative formula (2.4.4) (with $n = 2$), $\oint_{|z|=2} \frac{(z+1)e^z}{z^2} dz = 2\pi i f'(0) = 2\pi i(2)$.

Therefore

$$\oint_{|z|=2} \frac{e^z}{z^2(z-1)} dz = 2\pi i(e-2).$$

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Taylor series from contour integration

Theorem (Taylor Series)

A function $f(z)$ analytic in a neighborhood of z_0 has $f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} (z - z_0)^n f^{(n)}(z_0)$.

Proof. Start from Cauchy's formula and expand the kernel as a geometric series. With $|z - z_0| < |\xi - z_0|$,

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint \frac{f(\xi)}{\xi - z} d\xi = \frac{1}{2\pi i} \oint \frac{f(\xi)}{\xi - z_0} \frac{1}{1 - \frac{z - z_0}{\xi - z_0}} d\xi \\ &= \frac{1}{2\pi i} \oint \frac{f(\xi)}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n d\xi = \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z - z_0)^n \oint \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (z - z_0)^n f^{(n)}(z_0), \end{aligned}$$

using the derivative formula (2.4.4) in the last step. □

A function is analytic at z_0 if and only if it has a Taylor series there.

Taylor series from contour integration (cont.)

The geometric-series step uses $1 + r + r^2 + \cdots = \frac{1}{1-r}$ for $|r| < 1$, which follows from

$$(1-r)(1+r+\cdots+r^n) = 1 - r^{n+1} \xrightarrow{n \rightarrow \infty} 1.$$

Binomial expansion. For $f(z) = (1+z)^m$ with real m , the n th derivative at $z = 0$ is

$$\left. \frac{d^n}{dz^n} (1+z)^m \right|_{z=0} = m(m-1) \cdots (m-n+1),$$

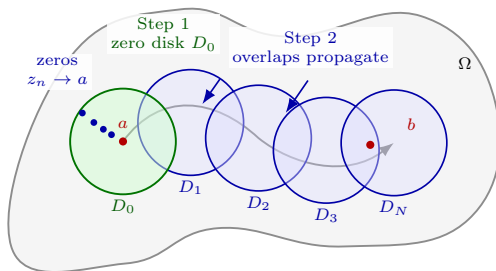
so

$$(1+z)^m = 1 + mz + \frac{m(m-1)}{1 \cdot 2} z^2 + \cdots = \sum_{n=0}^{\infty} \binom{m}{n} z^n. \quad (2.5.1)$$

If m is a non-negative integer the series terminates. Otherwise it is infinite, and since $(1+z)^{1/2} = e^{\frac{1}{2} \log(1+z)}$ has a branch point at $z = -1$, the expansion about 0 converges only for $|z| < 1$.

Analytic continuation

A general smooth function on $[0, 1]$ has no unique extension to $\mathbb{R}$, but a polynomial of degree n is fixed by $n + 1$ sampled points. A Taylor series likewise has a *unique* extension wherever continuation is possible: in $\mathbb{C}$, an analytic function has a unique *analytic continuation* along overlapping domains until a singularity or branch obstruction is met.



Example. The real graphs of $(1 + x^2)^{-1}$ and e^{-x^2} look alike, but their Taylor series differ: e^{-x^2} converges everywhere, while the Taylor series of $(1 + x^2)^{-1}$ about 0 has radius 1 and fails at $|x| = 1$. In the plane,

$$\frac{1}{1 + z^2} = \frac{1}{(i + z)(-i + z)}$$

has poles at $z = \pm i$, so the Taylor series about 0 converges only for $|z| < 1$. The function e^{-z^2} is entire, so its series is valid everywhere.

Theorem (Identity / splicing)

If f_1 and f_2 are analytic on a connected domain $U \cup V$, and if $f_1 = f_2$ on an overlap $U \cap V$ containing an accumulation point, then $f_1 = f_2$ on all of $U \cup V$.

Proof. Let $g = f_1 - f_2$. Then g is analytic on $U \cup V$ and $g = 0$ on $U \cap V$. Its Taylor series is identically zero there, and by uniqueness of analytic continuation $g \equiv 0$ on $U \cup V$. Hence $f_1 = f_2$. □

Laurent series

Near a singularity a *Laurent series* beats a Taylor series. It splits f into analytic and singular parts:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n = \underbrace{\sum_{n=0}^{\infty} a_n(z-z_0)^n}_{\text{analytic}} + \underbrace{\sum_{n=-\infty}^{-1} a_n(z-z_0)^n}_{\text{singular}}, \quad (2.5.2)$$

valid in an *annulus* about z_0 even when z_0 is singular. Multiplying by $(z-z_0)^{-k-1}$ and integrating term by term, only the $n=k$ term survives by (2.4.2), giving the coefficients

$$a_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z-z_0)^{n+1}} dz. \quad (2.5.3)$$

Laurent series (cont.)

Example (e^{-1/z^2}). On the real axis every derivative at 0 is zero, so the Taylor series is 0 everywhere—wrong off the origin, because $z = 0$ is an essential singularity. From the entire series for e^z ,

$$e^{-z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^{2n} \Rightarrow e^{-1/z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^{-2n},$$

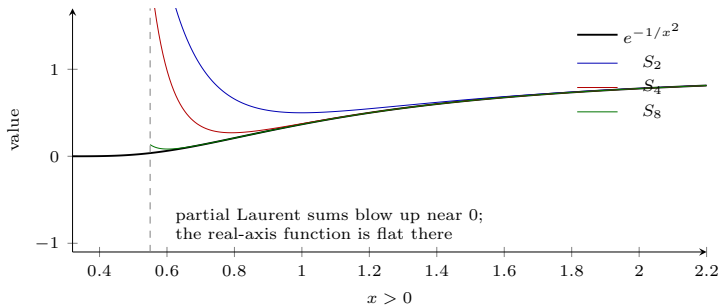
a Laurent series valid for all $z \neq 0$.

The real-axis flatness follows from the standard limit

$$\lim_{x \rightarrow 0} x^{-m} e^{-1/x^2} = 0 \quad (m = 0, 1, 2, \dots).$$

Thus $f(0) = 0$, $f'(0) = \lim_{x \rightarrow 0} e^{-1/x^2}/x = 0$, and the same argument applies after each differentiation: every derivative is a polynomial in $1/x$ times e^{-1/x^2} , so every derivative at 0 vanishes.

Laurent series (cont.)



Example (sinc). Since $\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$,

$$\operatorname{sinc} z = \frac{\sin z}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n}.$$

The singularity at $z = 0$ is removable: defining $\operatorname{sinc} 0 = 1$ makes it analytic.

Laurent series and the Bernoulli numbers

Example ($1/(e^z - 1)$). This has a simple pole at 0. Work with $g(z) = \frac{e^z - 1}{z} = \sum_{j=0}^{\infty} \frac{1}{(j+1)!} z^j$ and invert it. The *Cauchy product* of two power series is

$$\left(\sum_i a_i z^i \right) \left(\sum_j b_j z^j \right) = \sum_k c_k z^k, \quad c_k = \sum_{l=0}^k a_l b_{k-l},$$

so with $b_j = 1/(j+1)!$ and $c_0 = 1$, $c_k = 0$ ($k \geq 1$), the reciprocal coefficients satisfy $a_n = b_0^{-1} (c_n - \sum_{l=0}^{n-1} a_l b_{n-l})$. The first few are

$$a_0 = 1, \quad a_1 = -\frac{1}{2}, \quad a_2 = \frac{1}{12}, \quad a_3 = 0,$$

$$a_4 = -\left(\frac{1}{120} \cdot 1 + \frac{1}{24} \cdot \left(-\frac{1}{2}\right) + \frac{1}{6} \cdot \frac{1}{12} + \frac{1}{2} \cdot 0 \right) = -\frac{1}{720}.$$

Laurent series and the Bernoulli numbers (cont.)

Writing $a_n = B_n/n!$ defines the *Bernoulli numbers*

$$\{B_0, B_1, B_2, \dots\} = \{1, -\tfrac{1}{2}, \tfrac{1}{6}, 0, -\tfrac{1}{30}, 0, \tfrac{1}{42}, \dots\},$$

and dividing g^{-1} by z ,

$$\frac{1}{e^z - 1} = z^{-1} - \frac{1}{2} + \frac{1}{12}z - \frac{1}{720}z^3 + \dots = z^{-1} \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n. \quad (2.5.4)$$

The recurrence for the Bernoulli numbers (from Faulhaber's sum-of-powers formula) is

$$B_p = -\frac{1}{p+1} \sum_{j=0}^{p-1} B_j \binom{p+1}{j}. \quad (2.5.5)$$

Detour: Lovelace, Bernoulli numbers, and computation

Ada Lovelace's 1843 notes on Babbage's Analytical Engine include an algorithmic table for Bernoulli numbers. The computational point is exactly the recurrence just derived:

$$B_p = -\frac{1}{p+1} \sum_{j=0}^{p-1} \binom{p+1}{j} B_j.$$

Starting from $B_0 = 1$, every new value B_p depends only on earlier values $B_0, \dots, B_{p-1}$. Thus the recurrence is not merely a formal identity; it is an effective program for producing the coefficients in

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n.$$

This is the historical bridge in the book: special-function constants, power-series coefficients, and algorithms enter the story together.

Detour: Lovelace, Bernoulli numbers, and computation (cont.)

The same numbers appear in Faulhaber's formula. With the convention $B_1 = -1/2$,

$$\sum_{k=1}^n k^p = \frac{1}{p+1} \sum_{j=0}^p (-1)^j \binom{p+1}{j} B_j n^{p+1-j}. \quad (2.5.6)$$

For example, $p = 1$ gives

$$\sum_{k=1}^n k = \frac{1}{2}(n^2 + n).$$

Computing the needed binomial coefficients is itself recursive:

$$\binom{p+1}{j} = \binom{p}{j-1} + \binom{p}{j},$$

the Pascal-triangle rule used by Lovelace's tabular algorithm.

The Laurent series of $\cot z$

From $\cos z = \frac{1}{2}(e^{iz} + e^{-iz})$ and $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$,

$$\cot z = \frac{\cos z}{\sin z} = i \frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}}. \quad (2.5.7)$$

Splitting the fraction (note both terms keep the + sign),

$$\cot z = i \frac{e^{iz}}{e^{iz} - e^{-iz}} + i \frac{e^{-iz}}{e^{iz} - e^{-iz}} = i \left(\frac{1}{e^{2iz} - 1} + \frac{1}{1 - e^{-2iz}} \right).$$

Using (2.5.4) for $1/(e^z - 1)$ with $z \mapsto \pm 2iz$ and combining,

$$\cot z = i \sum_{p=0}^{\infty} \frac{B_p}{p!} \left[(2iz)^{p-1} - (-2iz)^{p-1} \right].$$

The bracket vanishes when p is odd, because then $p - 1$ is even; only the even Bernoulli indices $p = 2n$ remain. This gives

$$\cot z = z^{-1} \sum_{n=0}^{\infty} \frac{(-1)^n B_{2n} 2^{2n}}{(2n)!} z^{2n} = \frac{1}{z} - \frac{z}{3} - \frac{z^3}{45} - \dots. \quad (2.5.8)$$

(Check: $n = 0$ gives $1/z$; $n = 1$ gives $-B_2 4/2! \cdot z = -z/3$.)

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Classifying singularities

A *singularity* is a point where f is not defined. The Laurent series tells which kind:

- **Removable:** $a_n = 0$ for all $n < 0$. Example: $\operatorname{sinc} z$, with $\operatorname{sinc} 0 := 1$.
- **Pole of order p :** $a_{-p} \neq 0$ and $a_n = 0$ for $n < -p$. A simple pole has $p = 1$ (e.g. $e^z/(1-z)$ at $z = 1$); a double pole has $p = 2$ (e.g. $e^z/(1-z)^2$).
- **Essential:** infinitely many negative-power terms. Example:

$$\sin\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{-(2n+1)},$$

which is wild near the origin even though $\sin z$ is entire.

A function analytic except at isolated poles is called *meromorphic*; e.g. $1/\sin z$ is meromorphic with poles at $z = \pm n\pi$.

Classifying singularities (cont.)

Boundary language. A boundary point of a set S is a point that can be approached both from inside S and from outside S . An *open* set omits all its boundary points; a *closed* set contains them. This is why the ordinary interval $(0, 1)$ is open, $[0, 1]$ is closed, and the real line can be compactified by adding ideal endpoint(s) at infinity.

The Riemann sphere. Stereographic projection

$$z = \tan \frac{1}{2}\theta e^{i\phi}$$

maps the plane onto a sphere, adding a single point at infinity. This defines the *extended complex numbers* $\mathbb{C} \cup \{\infty\}$: the unit circle becomes the equator, the interior the southern hemisphere, and lines become circles passing through ∞ . A neighborhood of ∞ on the sphere corresponds to the exterior $\{|z| > R\}$ of a large disk in the plane.

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The residue theorem

The coefficient a_{-1} in the Laurent series is the *residue* of f at z_0 , written $\text{Res}[f, z_0]$. Integrating the Laurent series term by term and using $\oint_{\gamma} (z - z_0)^n dz = 2\pi i \delta_{n,-1}$, only the $n = -1$ term survives:

$$\oint_{\gamma} f(z) dz = 2\pi i a_{-1} = 2\pi i \text{Res}[f, z_0]. \quad (2.7.1)$$

For several poles, excluding each by a small loop (the function is analytic between them) gives the *residue theorem*:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_k \text{Res}[f, z_k] \quad (2.7.2)$$

summed over all poles z_k inside γ .

The residue theorem (cont.)

Computing residues.

- Simple pole: $\text{Res}[f, z_0] = \lim_{z \rightarrow z_0} (z - z_0)f(z)$.
- Double pole: $\text{Res}[f, z_0] = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0)^2 f(z)]$.
- Order- n pole: $\text{Res}[f, z_0] = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} [(z - z_0)^n f(z)]$.
- Ratio with simple zero of g : $\text{Res}\left[\frac{f}{g}, z_0\right] = \frac{f(z_0)}{g'(z_0)}$.
- At infinity: $\text{Res}[f, \infty] = -\text{Res}\left[\frac{1}{z^2} f(1/z), 0\right]$.

Examples.

$$\text{Res}\left[\frac{1}{z^2+1}, i\right] = \frac{1}{2z}\Big|_{z=i} = \frac{1}{2i}, \quad \text{Res}\left[\frac{1}{(z^2+1)^2}, i\right] = \frac{-2}{(z+i)^3}\Big|_{z=i} = \frac{1}{4i}.$$

$$\text{Res}\left[\frac{\sin z}{z^2}, 0\right] = \cos z\Big|_{z=0} = 1, \quad \text{Res}[z+1, \infty] = -\text{Res}\left[\frac{1}{z^3} + \frac{1}{z^2}, 0\right] = 0.$$

The residue theorem (cont.)

Two quick contour integrals. On $|z| = 1$,

$$\oint \frac{\sin z}{z^2} dz = 2\pi i \operatorname{Res}\left[\frac{\sin z}{z^2}, 0\right].$$

Since $\sin z = z - \frac{z^3}{3!} + \dots$, we have $\sin z/z^2 = z^{-1} - z/3! + \dots$, so the residue is 1 and the integral is $2\pi i$. For the other example,

$$\frac{z^2}{\sin z} = \frac{z^2}{z - \frac{z^3}{3!} + \dots} = \frac{z}{1 - \frac{z^2}{3!} + \dots},$$

which is analytic at 0 after defining its value there to be 0. Hence its residue is 0, and

$$\oint_{|z|=1} \frac{z^2}{\sin z} dz = 0.$$

Example ($z^2/(e^z - 1)$). The equation $e^z - 1 = 0$ gives $z = 2\pi im$ for $m \in \mathbb{Z}$. At $m = 0$, however, the numerator cancels the zero:

$$\frac{z^2}{e^z - 1} = \frac{z^2}{z + \frac{1}{2}z^2 + \dots} = \frac{z}{1 + \frac{1}{2}z + \dots},$$

The residue theorem (cont.)

so $z = 0$ is removable and its residue is 0. For $m \neq 0$ the poles are simple, and using $g(z) = e^z - 1$, $g'(z) = e^z$, gives

$$\operatorname{Res}\left[\frac{z^2}{e^z - 1}, 2\pi im\right] = \frac{(2\pi im)^2}{e^{2\pi im}} = -4\pi^2 m^2.$$

Branch-point example and a $\sum 1/n^2$ via residues

Branch point. For $f(z) = \frac{z^{-k}}{z+1}$ with $0 < k < 1$, use the branch $\arg z \in (0, 2\pi)$, with the cut along the positive real axis. Then $z = -1$ has $\arg z = \pi$, so the simple pole at $z = -1$ has residue

$$\operatorname{Res}[f, -1] = z^{-k} \Big|_{z=-1} = (-1)^{-k} = (e^{i\pi})^{-k} = e^{-i\pi k}.$$

The point $z = 0$ is a branch point, not an isolated singularity, so it has no Laurent series there (the pole/essential classification does not apply). A small circle of radius ε around 0 contributes at most $2\pi\varepsilon \cdot \varepsilon^{-k}/(1-\varepsilon)$, which tends to 0 as $\varepsilon \rightarrow 0$; this is a vanishing small-loop contribution, not a residue at 0.

Branch-point example and a $\sum 1/n^2$ via residues (cont.)

Basel sum $\sum_{n=1}^{\infty} \frac{1}{n^2}$. The function $f(z) = \frac{\pi \cot(\pi z)}{z^2}$ has poles at every integer. From the Laurent series of $\pi \cot \pi z$ about $z = n$,

$$\pi \cot \pi z = \frac{1}{z - n} - \frac{\pi^2(z - n)}{3} - \frac{\pi^4(z - n)^3}{45} - \dots, \quad (2.7.3)$$

one reads off $\text{Res}[f, n] = 1/n^2$ for $n \neq 0$ and $\text{Res}[f, 0] = -\pi^2/3$. On a square contour of side N (odd integer), the contour stays a distance at least $N/2$ from the origin and has length $4N$. Since $|\cot \pi z| < 2$ there, the *ML*-estimate gives

$$\left| \frac{1}{2\pi i} \oint f \, dz \right| \leq \frac{1}{2\pi} \cdot \frac{2\pi}{(N/2)^2} \cdot 4N = \frac{16}{N} \rightarrow 0.$$

Summing all residues to zero,

$$\left(\sum_{n=-\infty}^{-1} \frac{1}{n^2} \right) + \left(-\frac{\pi^2}{3} \right) + \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right) = 0 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

More generally $\sum_{n=1}^{\infty} \frac{1}{n^{2s}} = \frac{(2\pi)^{2s} |B_{2s}|}{2(2s)!}$, giving $\pi^4/90$ and $\pi^6/945$ for $s = 2, 3$.

Euler's Basel shortcut for the Basel problem

Euler's original route compares the zeros of $\sin \pi x$ with its Taylor coefficients. Using the infinite product

$$\sin \pi x = \pi x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right),$$

the x^3 term comes from choosing one factor $-x^2/n^2$ and taking 1 from all the others:

$$\sin \pi x = \pi x \left(1 - x^2 \sum_{n=1}^{\infty} \frac{1}{n^2} + O(x^4)\right) = \pi x - \pi x^3 \sum_{n=1}^{\infty} \frac{1}{n^2} + O(x^5).$$

The Taylor series is

$$\sin \pi x = \pi x - \frac{\pi^3 x^3}{3!} + O(x^5).$$

Comparing the x^3 coefficients gives $-\pi \sum_{n=1}^{\infty} n^{-2} = -\pi^3/6$, hence

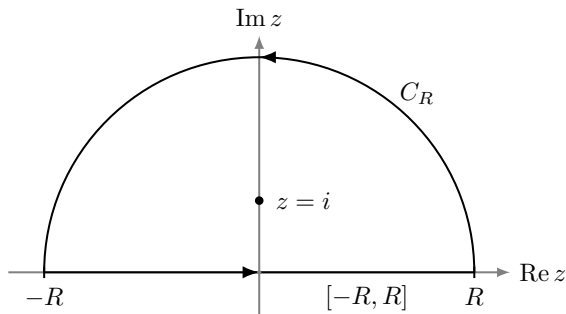
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

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Real integrals by a semicircular contour

To evaluate $\int_{-\infty}^{\infty} f(x) dx$, close the contour with a large semicircle in the upper half-plane:



Real integrals by a semicircular contour (cont.)

$$\underbrace{\oint_{\gamma} f \, dz}_{\textcircled{1}} = \underbrace{\int_{-R}^R f(x) \, dx}_{\textcircled{2}} + \underbrace{\int_0^{\pi} f(Re^{i\theta}) Re^{i\theta} i \, d\theta}_{\textcircled{3}}.$$

If the arc $\textcircled{3}$ vanishes as $R \rightarrow \infty$, then

$\int_{-\infty}^{\infty} f = \textcircled{1} = 2\pi i \sum (\text{residues in upper half-plane})$. The arc is controlled by the *ML-estimate*

$$\left| \int_{\gamma} f(z) \, dz \right| \leq ML, \quad M = \max_{z \in \gamma} |f(z)|, \quad L = \text{length of } \gamma,$$

which here gives $\left| \int_{C_R} f(z) \, dz \right| \leq \pi R \max_{C_R} |f| \rightarrow 0$ provided f decays faster than $1/R$.

Real integrals by a semicircular contour (cont.)

Example. $\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \operatorname{Re} \int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^2} dz$. Since $|e^{iz}| \leq 1$ in the upper half-plane, the arc obeys $|\int_{C_R}| \leq \frac{1}{R^2-1} \pi R \rightarrow 0$. The pole at $z = i$ gives

$$\oint \frac{e^{iz}}{1+z^2} dz = 2\pi i \left. \frac{e^{iz}}{2z} \right|_{z=i} = 2\pi i \frac{e^{-1}}{2i} = \pi e^{-1},$$

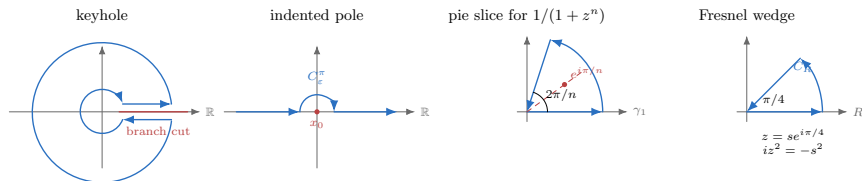
$$\text{so } \int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{e}.$$

Example. $\int_{-\infty}^{\infty} \frac{1}{1+x^4} dx$. The roots of $1+z^4$ are $e^{i\pi/4}, e^{i3\pi/4}, \dots$; two lie in the upper half-plane. Summing residues $\frac{1}{4z^3}$ there,

$$\oint \frac{dz}{1+z^4} = 2\pi i \left(\frac{1}{4} e^{-i3\pi/4} + \frac{1}{4} e^{-i9\pi/4} \right) = \frac{\pi}{\sqrt{2}},$$

$$\text{hence } \int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{\sqrt{2}}.$$

Contour atlas used below



The keyhole contour compares the two sides of a branch cut; the indented contour isolates a real-axis pole; the pie-slice contour matches a rotated ray to the real-axis integral; the Fresnel wedge rotates e^{iz^2} into a Gaussian.

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$

Evaluate $\int_0^\infty \frac{x^a}{(1+x)^2} dx$ for $-1 < a < 1$. First assume $a \neq 0$; the value at $a = 0$ follows by continuity. The factor $z^a = |z|^a e^{ia \arg z}$ is multivalued, so use the branch $0 < \arg z < 2\pi$, cut along the positive real axis and use a *keyhole* contour. The integrand has a double pole at $z = -1$:

$$\oint_C \frac{z^a}{(1+z)^2} dz = 2\pi i \left. \frac{d}{dz}(z^a) \right|_{z=-1} = 2\pi i a e^{i\pi(a-1)} = -2\pi i a e^{i\pi a}.$$

The large circle has length $2\pi R$ and integrand size $O(R^a/R^2) = O(R^{a-2})$, so its contribution is $O(R^{a-1}) \rightarrow 0$ because $a < 1$. The small circle has length $2\pi\epsilon$ and integrand size $O(\epsilon^a)$, so its contribution is $O(\epsilon^{a+1}) \rightarrow 0$ because $a > -1$. Above the cut $z^a = |z|^a$; below it $z^a = |z|^a e^{i2\pi a}$, so the two straight pieces combine to $(1 - e^{i2\pi a}) \int_0^\infty \frac{x^a}{(1+x)^2} dx$.

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

Therefore

$$(1 - e^{i2\pi a}) \int_0^\infty \frac{x^a}{(1+x)^2} dx = -2\pi i a e^{i\pi a}.$$

Dividing and writing $1 - e^{i2\pi a} = e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) = e^{i\pi a}(-2i \sin \pi a)$,

$$\int_0^\infty \frac{x^a}{(1+x)^2} dx = \frac{-2\pi i a e^{i\pi a}}{e^{i\pi a}(-2i \sin \pi a)} = \frac{\pi a}{\sin \pi a}. \quad (2.8.1)$$

The companion integral uses the same branch cut, but it is short enough to compute here. For $f(z) = z^{-a}/(1+z)$ with $0 < a < 1$, the pole at $z = -1$ has residue $(-1)^{-a} = e^{-i\pi a}$ on the chosen branch. The upper and lower straight pieces give $(1 - e^{-i2\pi a}) \int_0^\infty x^{-a}/(1+x) dx$. Hence

$$(1 - e^{-i2\pi a}) \int_0^\infty \frac{x^{-a}}{1+x} dx = 2\pi i e^{-i\pi a}.$$

Since $1 - e^{-i2\pi a} = e^{-i\pi a}(e^{i\pi a} - e^{-i\pi a}) = 2ie^{-i\pi a} \sin(\pi a)$,

$$\int_0^\infty \frac{x^{-a}}{1+x} dx = \frac{\pi}{\sin \pi a}, \quad 0 < a < 1. \quad (2.8.2)$$

Cauchy principal value and a log integral

When a pole lies on the path of integration, the ordinary integral diverges. The *Cauchy principal value* lets both sides approach the pole at the same rate. For example, the two one-sided integrals of $1/x$ on $[-1, 1]$ diverge, but

$$\text{PV} \int_{-1}^1 \frac{dx}{x} = \lim_{\varepsilon \rightarrow 0^+} \left(\int_{-1}^{-\varepsilon} \frac{dx}{x} + \int_{\varepsilon}^1 \frac{dx}{x} \right) = 0.$$

If the two sides are not synchronized the cancellation can change, so this is a separate notion from ordinary improper convergence:

$$\text{PV} \int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \left(\int_a^{x_0 - \varepsilon} + \int_{x_0 + \varepsilon}^b \right) f(x) dx.$$

Indenting the contour by a small semicircle C_ε^θ of angle θ around a simple pole contributes

$$\int_{C_\varepsilon^\theta} f(z) dz \rightarrow i\sigma\theta \text{Res}[f, z_0], \quad (2.8.3)$$

a fraction $\theta/2\pi$ of a full residue. Here $\sigma = +1$ for counterclockwise indentation and $\sigma = -1$ for clockwise indentation. For a half-circle, $\theta = \pi$.

Cauchy principal value and a log integral (cont.)

Example. $\int_0^\infty \frac{\log x}{x^2 - 1} dx$. Use $f(z) = \frac{\text{Log } z}{z^2 - 1}$ on an upper semicircle indented at $z = -1, 0, 1$, with the principal boundary value $\text{Log}(-1 + i0) = i\pi$. The big and small arcs vanish. At $z = -1$ (a half-circle clockwise, $\theta = \pi$),

$$\int_{C_{-1}} \frac{\log z}{z^2 - 1} dz = -\pi i \frac{\log z}{2z} \Big|_{z=-1} = -\pi i \frac{i\pi}{-2} = -\frac{\pi^2}{2},$$

while at $z = 1$ the residue $\log 1 = 0$ gives no contribution. Collecting real parts (the $\int_0^\infty$ piece is half the symmetric integral) yields

$$0 = \text{PV} \int_{-\infty}^\infty \frac{\text{Log}(x + i0)}{x^2 - 1} dx + \left(-\frac{\pi^2}{2}\right) + 0,$$

so

$$\text{PV} \int_{-\infty}^\infty \frac{\text{Log}(x + i0)}{x^2 - 1} dx = \frac{\pi^2}{2}, \quad \int_0^\infty \frac{\log x}{x^2 - 1} dx = \frac{\pi^2}{4}.$$

Worked example: a squared-log integral

Evaluate

$$I = \int_0^{\infty} \frac{(\log x)^2}{1+x^2} dx.$$

Use the principal logarithm in the upper half-plane, so $\text{Log } x = \log x$ for $x > 0$ and $\text{Log}(-x + i0) = \log x + i\pi$ for $x > 0$. Consider

$$F(z) = \frac{\text{Log}^2 z}{1+z^2}$$

on the upper semicircle. The large arc vanishes because $|\text{Log}^2 z/(1+z^2)| = O((\log R)^2/R^2)$ while the arc length is πR ; the small indentation around 0 vanishes because $\varepsilon(\log \varepsilon)^2 \rightarrow 0$.

Worked example: a squared-log integral (cont.)

Only the pole at $z = i$ lies inside the contour. Since $1 + z^2 = (z - i)(z + i)$,

$$\operatorname{Res}[F, i] = \frac{\operatorname{Log}^2 i}{2i} = \frac{(i\pi/2)^2}{2i} = \frac{-\pi^2/4}{2i} = \frac{i\pi^2}{8}.$$

Therefore

$$\operatorname{PV} \int_{-\infty}^{\infty} \frac{\operatorname{Log}^2(x + i0)}{1 + x^2} dx = 2\pi i \frac{i\pi^2}{8} = -\frac{\pi^3}{4}.$$

Worked example: a squared-log integral (cont.)

Split the real-axis integral into positive and negative halves. On $x > 0$ the integrand is $(\log x)^2/(1+x^2)$. On $x < 0$, write $x = -t$:

$$\operatorname{Log}(-t + i0) = \log t + i\pi,$$

so the negative half contributes

$$\int_0^\infty \frac{(\log t + i\pi)^2}{1+t^2} dt = I + 2i\pi \int_0^\infty \frac{\log t}{1+t^2} dt - \pi^2 \int_0^\infty \frac{dt}{1+t^2}.$$

The middle integral is zero by the substitution $t = 1/u$, and the last integral is $\pi/2$. Thus the full contour value is

$$I + \left(I - \frac{\pi^3}{2}\right) = 2I - \frac{\pi^3}{2}.$$

Equating with $-\pi^3/4$ gives

$$I = \int_0^\infty \frac{(\log x)^2}{1+x^2} dx = \frac{\pi^3}{8}. \quad (2.8.4)$$

Jordan's lemma and $\int \sin x/x \, dx$

Theorem (Jordan's Lemma)

If f is analytic in the upper half-plane except at finitely many singularities and $|f| \rightarrow 0$ as $|z| \rightarrow \infty$ there, then $\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{iaz} dz = 0$ for $a > 0$.

Proof sketch. With $z = Re^{i\theta}$ and $M_R = \max_{C_R} |f|$,

$$\left| \int_{C_R} f e^{iaz} dz \right| \leq M_R R \int_0^\pi e^{-aR \sin \theta} d\theta = 2M_R R \int_0^{\pi/2} e^{-aR \sin \theta} d\theta.$$

Using $\sin \theta \geq 2\theta/\pi$ on $[0, \pi/2]$ and integrating the exponential,

$$\leq 2M_R R \left(\frac{-\pi}{2aR} \right) e^{-2aR\theta/\pi} \Big|_0^{\pi/2} = M_R \frac{\pi}{a} (1 - e^{-aR}) \rightarrow 0,$$

since $M_R \rightarrow 0$. □

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

Example. $\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx$. Take the imaginary part of $\int \frac{e^{iz}}{z} \, dz$ on a semicircle indented at $z = 0$. The big arc vanishes by Jordan's lemma; the small indentation about the pole contributes, in the limit as $\varepsilon \rightarrow 0$,

$$\lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} \frac{e^{iz}}{z} \, dz = -\pi i \operatorname{Res}\left[\frac{e^{iz}}{z}, 0\right] = -\pi i.$$

With no poles enclosed, the full contour is zero, so $\operatorname{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} \, dx = \pi i$, and taking the imaginary part,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \pi.$$

Pie-slice contour: $\int_0^\infty 1/(1+x^n) dx$

For an integer $n > 1$, the zeros of $1+z^n$ are $z = e^{i(2k+1)\pi/n}$; only $z = e^{i\pi/n}$ lies inside a wedge of angle $2\pi/n$. The contour is the boundary of that wedge: γ_1 out along the real axis, arc C_R , and γ_2 back at angle $2\pi/n$. At the simple pole the residue is $1/(nz^{n-1})$. Since $z_0^n = e^{i\pi} = -1$ there, $z_0^{n-1} = z_0^n/z_0 = -e^{-i\pi/n}$, so $1/(nz_0^{n-1}) = -e^{i\pi/n}/n$ and

$$\oint_{\gamma} \frac{dz}{1+z^n} = 2\pi i \frac{1}{nz^{n-1}} \Big|_{z=e^{i\pi/n}} = -\frac{2\pi i}{n} e^{i\pi/n}.$$

Pie-slice contour: $\int_0^\infty 1/(1+x^n) dx$ (cont.)

On the return ray $z = re^{i2\pi/n}$ gives $z^n = r^n$ and $dz = e^{i2\pi/n} dr$, so γ_2 reproduces the real integral times $-e^{i2\pi/n}$. With the arc vanishing,

$$(1 - e^{i2\pi/n}) \int_0^\infty \frac{dr}{1+r^n} = -\frac{2\pi i}{n} e^{i\pi/n}.$$

Dividing and writing the denominator as $1 - e^{i2\pi/n} = e^{i\pi/n}(e^{-i\pi/n} - e^{i\pi/n})$, the factor $e^{i\pi/n}$ cancels the one in the numerator; then $e^{-i\pi/n} - e^{i\pi/n} = -2i \sin(\pi/n)$ by (2.2.4), so

$$\int_0^\infty \frac{dx}{1+x^n} = -\frac{2\pi i}{n} \frac{e^{i\pi/n}}{e^{i\pi/n}(e^{-i\pi/n} - e^{i\pi/n})} = -\frac{2\pi i}{n} \frac{1}{-2i \sin(\pi/n)} = \frac{\pi/n}{\sin(\pi/n)}. \quad (2.8.5)$$

sinc transform, Gaussian, and Fresnel integrals

Fourier transform of sinc. Writing $\frac{\sin t}{t} = \frac{e^{it} - e^{-it}}{2it}$ and combining with $e^{i\xi t}$ gives

$$\frac{\sin t}{t} e^{i\xi t} = \frac{e^{i(\xi+1)t} - e^{i(\xi-1)t}}{2it}.$$

Jordan's lemma gives the principal-value contour rule $\int_{-\infty}^{\infty} e^{iat} t^{-1} dt = i\pi \operatorname{sgn}(a)$, so the transform is $\frac{\pi}{2} [\operatorname{sgn}(\xi + 1) - \operatorname{sgn}(\xi - 1)]$, with half-height at the jump points. Thus

$$\int_{-\infty}^{\infty} \frac{\sin t}{t} e^{i\xi t} dt = \begin{cases} \pi, & -1 < \xi < 1, \\ \pi/2, & |\xi| = 1, \\ 0, & |\xi| > 1. \end{cases} \quad (2.8.6)$$

Gaussian integral. Squaring and going to polar coordinates,

$$\left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = \pi,$$

so $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ and, scaling $x \mapsto x/\sqrt{m}$,

$$\int_{-\infty}^{\infty} e^{-mx^2} dx = \sqrt{\pi/m}, \quad m > 0. \quad (2.8.7)$$

sinc transform, Gaussian, and Fresnel integrals (cont.)

Fresnel integrals. Take the real and imaginary parts of $\int_0^\infty e^{iz^2} dz$ on a pie slice of angle $\pi/4$. On the outer arc $z = Re^{i\theta}$, $0 \leq \theta \leq \pi/4$, substitute $u = z^2$. Its image is a quarter-circle in the upper half u -plane, with the branch $\sqrt{u} = z$ on that arc and $dz = du/(2\sqrt{u})$. Since $|\sqrt{u}| = R$ there, Jordan's lemma gives

$$\left| \int_{C_R} e^{iz^2} dz \right| = \left| \int_{\Gamma_{R^2}} \frac{e^{iu}}{2\sqrt{u}} du \right| \leq \frac{C}{R} \rightarrow 0,$$

where Γ_{R^2} is the image quarter-circle and the last estimate is Jordan's lemma on that arc, with C independent of R . Along the ray $z = e^{i\pi/4}s$ the integral reduces to a Gaussian:

$$\int_0^\infty e^{iz^2} dz = e^{i\pi/4} \int_0^\infty e^{-s^2} ds = e^{i\pi/4} \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{2} \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right).$$

Hence

$$\int_0^\infty \cos(x^2) dx = \int_0^\infty \sin(x^2) dx = \frac{\sqrt{\pi}}{2\sqrt{2}}. \quad (2.8.8)$$

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The steepest-descent approximation

We estimate

$$I(s) = \int_a^b g(z) e^{sf(z)} dz, \quad s \gg 1, \quad (2.9.1)$$

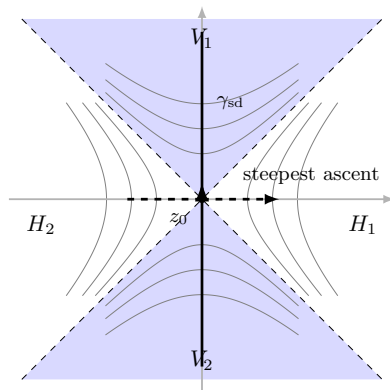
with f, g analytic. Write $f = u + iv$. At a saddle $f'(z_0) = 0$, so $u_x = v_x = 0$ at z_0 ; the Cauchy–Riemann equations then also give $u_y = -v_x = 0$ and $v_y = u_x = 0$. Thus both u and v have stationary first-order behavior there.

Differentiating the Cauchy–Riemann equations gives $u_{xx} + u_{yy} = 0$. The Hessian of $u = \operatorname{Re} f$ at z_0 has the form

$$D^2 u(z_0) = \begin{pmatrix} u_{xx} & u_{xy} \\ u_{xy} & -u_{xx} \end{pmatrix}, \quad \det D^2 u(z_0) = -u_{xx}^2 - u_{xy}^2.$$

If this Hessian is nonzero, the determinant is negative, so u has a genuine saddle: increase in one direction and decrease in another.

The steepest-descent approximation (cont.)



The steepest-descent approximation (cont.)

Assume the contour can be deformed without crossing singularities or branch cuts to a nondegenerate saddle z_0 where $f'(z_0) = 0$ and $f''(z_0) \neq 0$, and that $\operatorname{Re} f$ decreases along the chosen path away from the saddle. Near z_0 ,

$$f(z) \approx f(z_0) + \frac{1}{2}f''(z_0)(z - z_0)^2.$$

Write $f''(z_0) = |f''(z_0)|e^{i\theta}$ with $\theta = \arg f''(z_0)$, and take the path $z = z_0 + re^{i\phi}$. Choosing the *steepest-descent* direction $\phi = \frac{1}{2}(\pi - \theta)$ turns the exponent into a real Gaussian:

$$e^{\frac{1}{2}sf''(z_0)(z-z_0)^2} = e^{\frac{1}{2}s|f''(z_0)|e^{i\theta}r^2e^{i2\phi}} = e^{-\frac{1}{2}s|f''(z_0)|r^2}.$$

The steepest-descent approximation (cont.)

For $s \gg 1$ only the neighborhood of z_0 matters; set $g \approx g(z_0)$ and extend the Gaussian over all r . We parametrize the chosen descent direction by $z = z_0 + ie^{-i\theta/2}r$ with r increasing along the deformed contour, so $dz = ie^{-i\theta/2}dr$; reversing the contour would reverse this sign.

$$I(s) \approx g(z_0) e^{sf(z_0)} i e^{-i\theta/2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}s|f''(z_0)|r^2} dr,$$

and using the Gaussian (2.8.7),

$$I(s) \approx g(z_0) e^{sf(z_0)} i e^{-i\theta/2} \left(\frac{2\pi}{s|f''(z_0)|} \right)^{1/2} = \frac{\sqrt{2\pi} g(z_0) e^{sf(z_0)} i e^{-i\theta/2}}{\sqrt{s|f''(z_0)|}}. \quad (2.9.2)$$

Worked example: large binomial coefficient

Approximate $\binom{n}{m} = \frac{n!}{m!(n-m)!}$ for large n , with $0 < m < n$, $t = m/n \in (0, 1)$, and both m and $n - m$ large. From the binomial expansion $(1+z)^n = \sum_k \binom{n}{k} z^k$, the coefficient is a contour integral:

$$\binom{n}{m} = \frac{1}{2\pi i} \oint \frac{(1+z)^n}{z^{m+1}} dz = \frac{1}{2\pi i} \oint \frac{1}{z} e^{n[\log(1+z) - t \log z]} dz, \quad t = \frac{m}{n}.$$

So $f(z) = \log(1+z) - t \log z$. Its derivatives:

$$f'(z) = \frac{1}{1+z} - \frac{t}{z}, \quad f''(z) = -\frac{1}{(1+z)^2} + \frac{t}{z^2}.$$

Setting $f'(z_0) = 0$ gives the saddle $z_0 = \frac{t}{1-t} = \frac{m}{n-m}$, real and positive, with

$$f''(z_0) = \frac{(1-t)^3}{t}, \text{ so } \theta = \arg f''(z_0) = 0.$$

Worked example: large binomial coefficient (cont.)

Here the path through the (real) saddle goes in the imaginary direction, $\phi = \frac{1}{2}\pi$, which contributes a phase $i = e^{i\pi/2}$. Applying (2.9.2) with $g(z) = 1/(2\pi iz)$,

$$\binom{n}{m} \approx \frac{1}{2\pi i} \frac{1}{z_0} e^{i\pi/2} e^{nf(z_0)} \left(\frac{2\pi}{nf''(z_0)} \right)^{1/2} = \frac{\exp[-n(t \log t + (1-t) \log(1-t))]}{\sqrt{2\pi nt(1-t)}}. \quad (2.9.3)$$

Equivalently $\binom{n}{m} \approx \frac{1}{\sqrt{2\pi}} \frac{n^{n+\frac{1}{2}}}{m^{m+\frac{1}{2}}(n-m)^{n-m+\frac{1}{2}}}.$

Numerical check. For $n = 88$, $m = 18$ the exact value is

$$\binom{88}{18} = 2418561960739869780 \approx 2.4186 \times 10^{18},$$

and (2.9.3) gives 2.4304×10^{18} , within half a percent.

Summary

- A function is **analytic** on an open neighborhood where its complex derivative exists. With continuous first partials on that neighborhood, the **Cauchy–Riemann conditions** are equivalent to complex differentiability; the resulting analytic function has a convergent **Taylor series** locally.
- Near a singularity the **Laurent series** applies; its a_{-1} coefficient is the **residue**.
- **Cauchy's theorem** ($\oint f = 0$ for analytic f) and the **residue theorem** ($\oint f = 2\pi i \sum \text{Res}$) evaluate contour integrals by counting poles.
- Closing contours (semicircle, keyhole, pie-slice, indented) plus **Jordan's lemma** and the **ML-estimate** compute real integrals and infinite sums ($\sum 1/n^2 = \pi^2/6$, the Gaussian and Fresnel integrals, $\pi a / \sin \pi a$).
- The **method of steepest descent** gives large-parameter asymptotics, e.g. the binomial coefficient—the tool we carry into the study of the named special functions.